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```
clc;
close all;
clear all;
```

parameters

```
Fs=5000;                        % sampling frequency
T=1;                            % simulation time
machine_RPM=1800;
f_rot=machine_RPM/60;           % rotational frequency

A=1;
H=0.4;                          % 2nd harmonic amplitude
k=0.8;                          % transmission factor
sigma=0.15;                     % random noise

f_electrical=50;                % electrical interference
A_electrical=0.25;

f_high=300;                     % high frequency interference
A_high=0.20;

notch_low=48;
notch_high=52;
fc=100;                         % low pass cutoff

t=0:1/Fs:T;
L=length(t);
freq=(0:L-1)*(Fs/L);
```

signal

```
x1=A*sin(2*pi*f_rot*t);                 % fundamental
x2=H*sin(2*pi*2*f_rot*t);               % 2nd harmonic
```

```
x=x1+x2;                                    % original vibration

a_x=k*x;                                    % attenuation

electrical_noise=A_electrical*sin(2*pi*f_electrical*t); % 50 Hz
high_freq_noise=A_high*sin(2*pi*f_high*t);               % 300 Hz
random_noise=sigma*randn(size(t));                       % random noise

noise=electrical_noise+high_freq_noise+random_noise;
r_x=a_x+noise;                               % received signal
```

calculation

```
x_rms=rms(x);
x_peak=max(abs(x));

x_power=rms(a_x)^2;
noise_power=rms(noise)^2;
SNR=10*log10(x_power/noise_power);

[bn,an]=butter(4,[notch_low notch_high]/(Fs/2),'stop'); % remove 50 Hz
notch_signal=filtfilt(bn,an,r_x);

[bl,al]=butter(4,fc/(Fs/2),'low');          % remove high frequencies
filtered_signal=filtfilt(bl,al,notch_signal);

recovered_signal=filtered_signal/k;         % compensate attenuation

rx_rms=rms(r_x);
rx_peak=max(abs(r_x));
rx_power=rms(r_x)^2;
rx_snr=10*log10(rms(a_x)^2/rms(noise)^2);

filtered_rms=rms(recovered_signal);
filtered_peak=max(abs(recovered_signal));
filtered_power=rms(recovered_signal)^2;
```

plotting

```
% Figure 1 - transmission
figure();
plot(t,x,'r');
xlabel('Time (s)');
ylabel('Amplitude');
title('Transmission Signal');
xlim([0,0.2]);
grid on;

% Figure 2 - signal journey
figure();
subplot(3,1,1);
plot(t,x,'b');
xlabel('Time (s)');
```

```
ylabel('Amplitude');
title('Original Signal');
xlim([0,0.2]);
grid on;

subplot(3,1,2);
plot(t,a_x,'r');
xlabel('Time (s)');
ylabel('Amplitude');
title('Attenuated Signal');
xlim([0,0.2]);
grid on;

subplot(3,1,3);
plot(t,r_x,'k');
xlabel('Time (s)');
ylabel('Amplitude');
title('Received Signal');
xlim([0,0.2]);
grid on;

% Figure 3 - receiver filtering
figure;

subplot(3,1,1);
plot(t, r_x);
xlabel('Time (s)');
ylabel('Amplitude');
title('Received Noisy Signal');
grid on;
xlim([0 0.2]);

subplot(3,1,2);
plot(t, notch_signal);
xlabel('Time (s)');
ylabel('Amplitude');
title('After 50 Hz Band-Stop Filter');
grid on;
xlim([0 0.2]);

subplot(3,1,3);
plot(t, filtered_signal,'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Amplitude');
title('After 100 Hz Low-Pass Filter');
grid on;
xlim([0 0.2]);

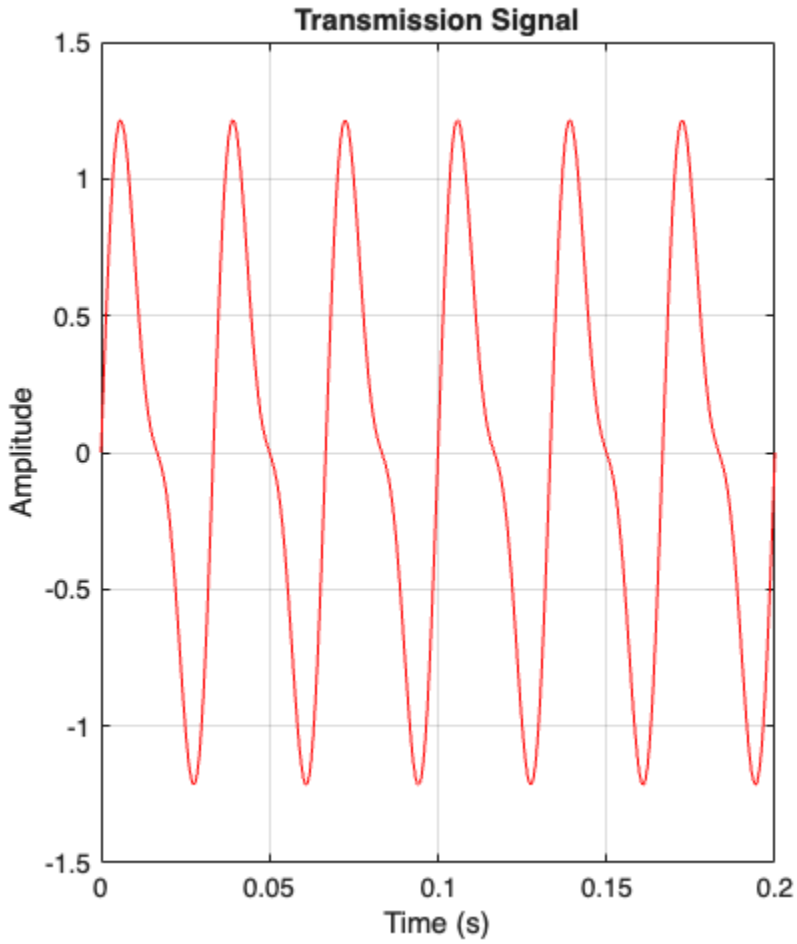
% Figure 4 - comparison
figure();
plot(t,x,'r',linewidth=1);
hold on;
plot(t,recovered_signal,'b',linewidth=1);
xlabel('Time (s)');
```

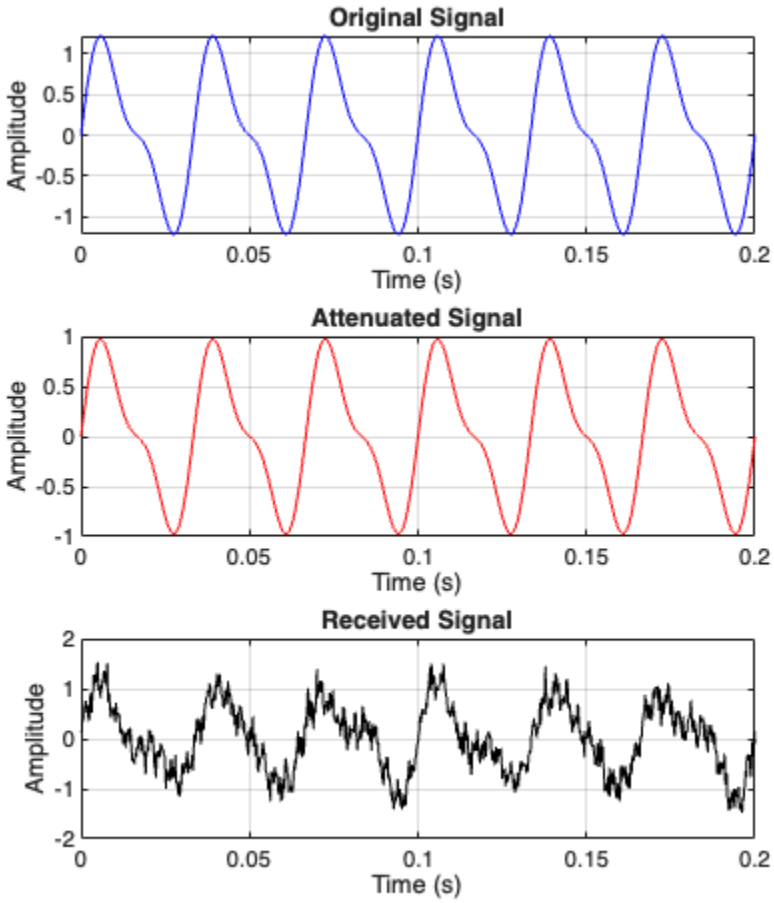
```
ylabel('Amplitude');
title('Original vs Filtered Signal');
legend('Original','Filtered');
xlim([0,0.2]);
grid on;

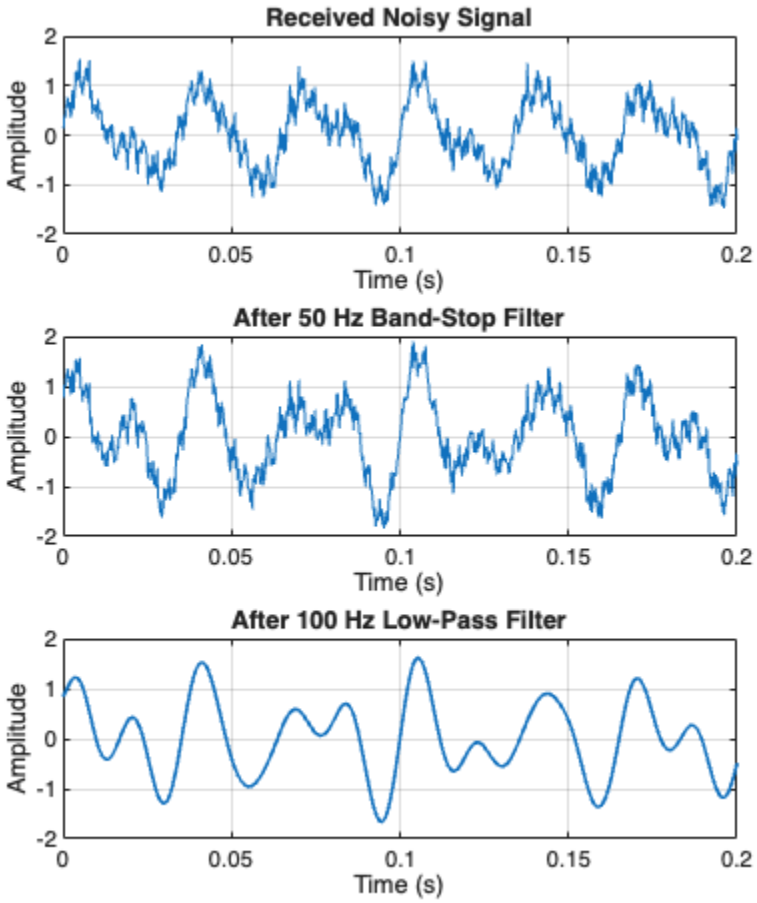
% Figure 5 - FFT analysis
F_x=fft(x);
F_rec=fft(r_x);
F_filtered=fft(recovered_signal);

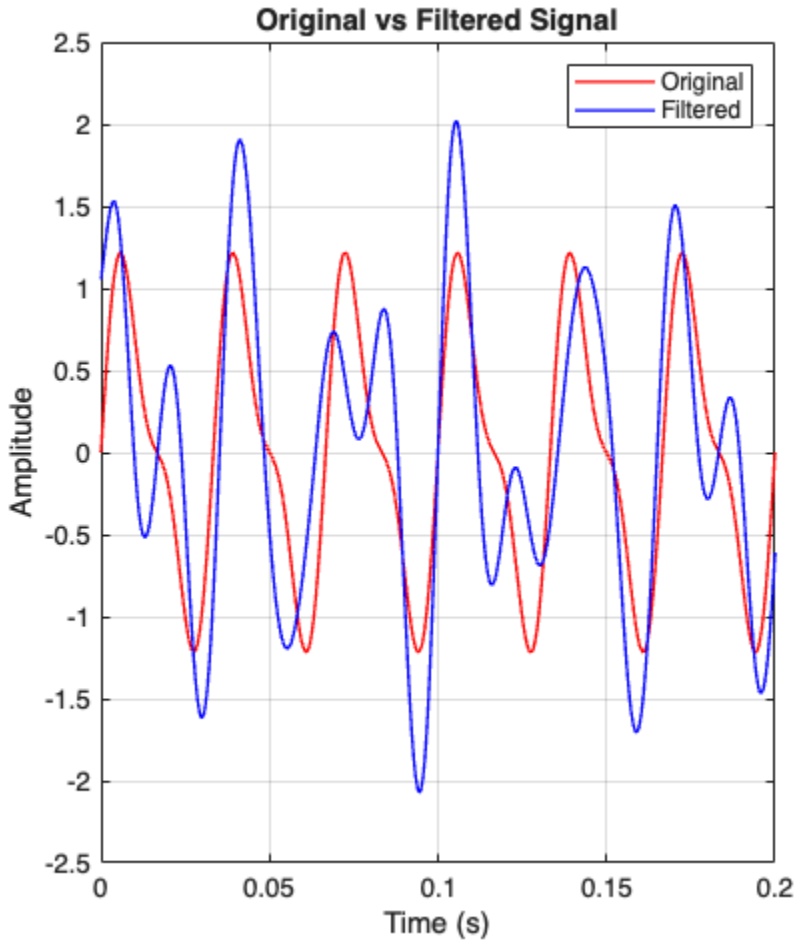
P_x=abs(F_x/L);
P_rec=abs(F_rec/L);
P_filtered=abs(F_filtered/L);

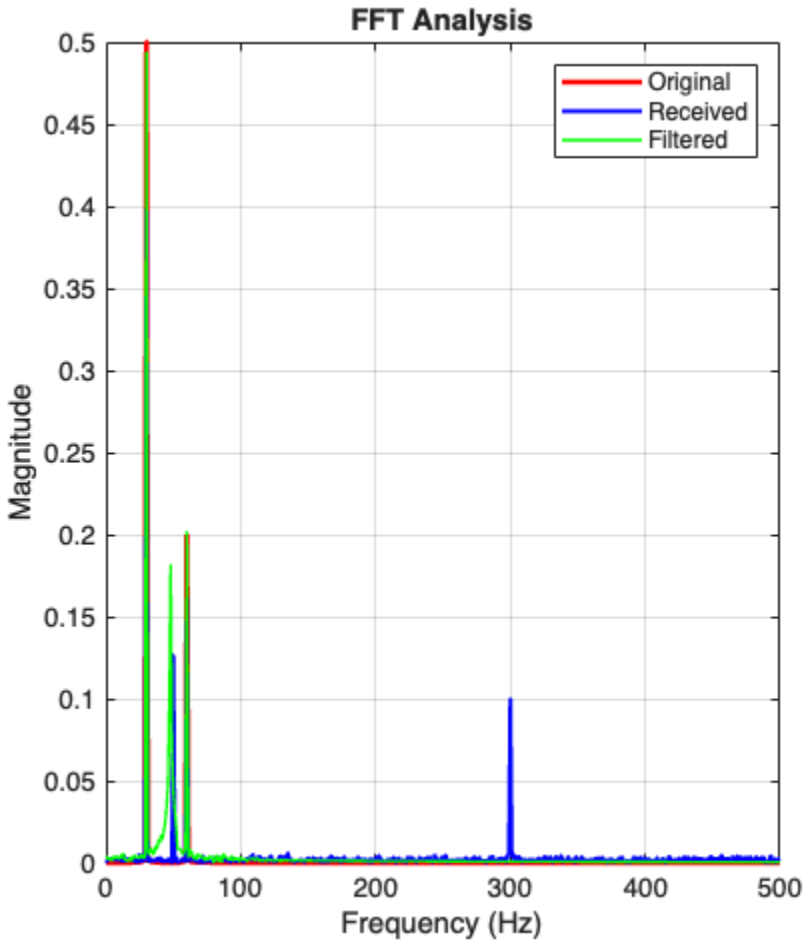
figure();
plot(freq(1:floor(L/2)),P_x(1:floor(L/2)),'r','LineWidth',2);
hold on;
plot(freq(1:floor(L/2)),P_rec(1:floor(L/2)),'b','LineWidth',1.5);
plot(freq(1:floor(L/2)),P_filtered(1:floor(L/2)),'g','LineWidth',1);
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('FFT Analysis');
legend('Original','Received','Filtered');
xlim([0 500]);
grid on;
```


Transmission Signal
1.5
1
0.5
0
-0.5
-1
-1.5
Amplitude
0
0.05
0.1
0.15
0.2
Time (s)


Original Signal
Amplitude
Time (s)
Attenuated Signal
Amplitude
Time (s)
Received Signal
Amplitude
Time (s)


Received Noisy Signal
Amplitude
Time (s)
After 50 Hz Band-Stop Filter
Amplitude
Time (s)
After 100 Hz Low-Pass Filter
Amplitude
Time (s)


Original vs Filtered Signal
Original
Filtered
Amplitude
Time (s)
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
0
0.05
0.1
0.15
0.2



display

```
fprintf('Parameters\n');
fprintf('Machine speed: %0.0f RPM\n',machine_RPM);
fprintf('Rotational frequency: %0.2f Hz\n',f_rot);
fprintf('Sampling frequency: %0.0f Hz\n',Fs);
fprintf('Simulation time: %0.2f sec\n',T);
fprintf('Attenuation factor: %0.2f\n',k);
fprintf('Noise STD: %0.2f\n',sigma);

fprintf('\nValues:\n');
fprintf('Original RMS: %0.4f\n',x_rms);
fprintf('Original Peak: %0.4f\n',x_peak);

fprintf('\nReceived signal:\n');
fprintf('RMS: %0.4f\n',rx_rms);
fprintf('Peak: %0.4f\n',rx_peak);
fprintf('Power: %0.4f\n',rx_power);
fprintf('SNR: %0.4f dB\n',rx_snr);
```

```
fprintf('\nFiltered signal:\n');
fprintf('RMS: %0.4f\n',filtered_rms);
fprintf('Peak: %0.4f\n',filtered_peak);
fprintf('Power: %0.4f\n',filtered_power);
```

```
Parameters
Machine speed: 1800 RPM
Rotational frequency: 30.00 Hz
Sampling frequency: 5000 Hz
Simulation time: 1.00 sec
Attenuation factor: 0.80
Noise STD: 0.15

Values:
Original RMS: 0.7615
Original Peak: 1.2147

Received signal:
RMS: 0.6661
Peak: 1.6240
Power: 0.4436
SNR: 6.9514 dB

Filtered signal:
RMS: 0.8293
Peak: 2.0686
Power: 0.6877
```

My reference for ppt

```
figure();
plot(t,x,'r--',linewidth=1);
hold on;
plot(t,r_x,'b');

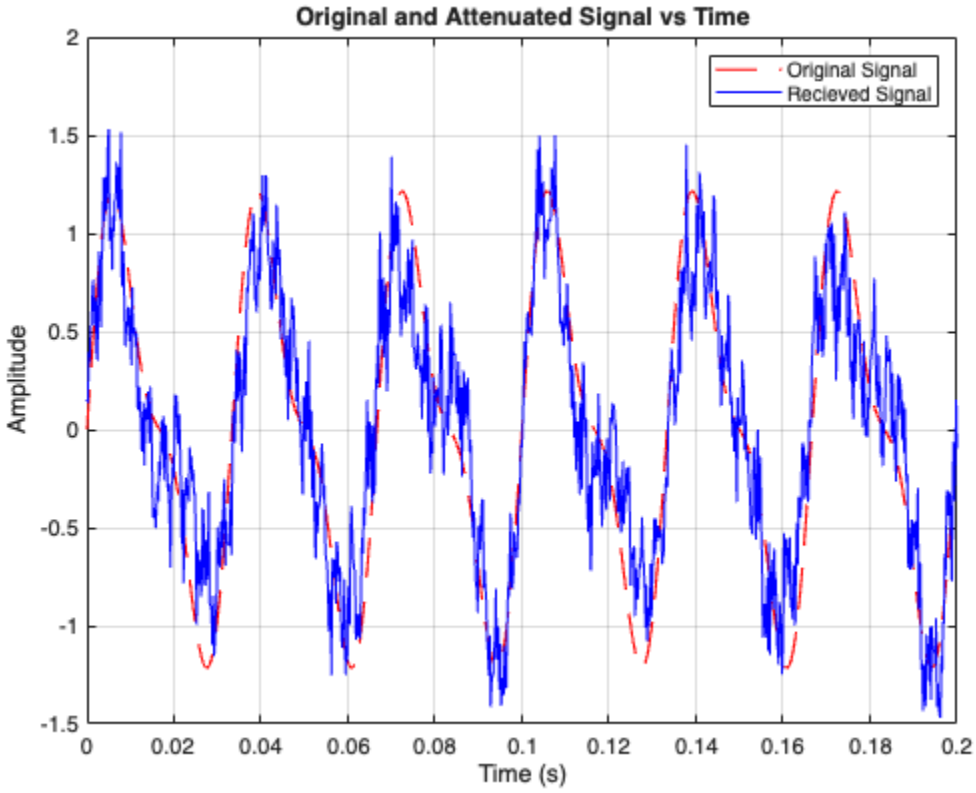
xlabel('Time (s)');
ylabel('Amplitude');
title('Original and Attenuated Signal vs Time');

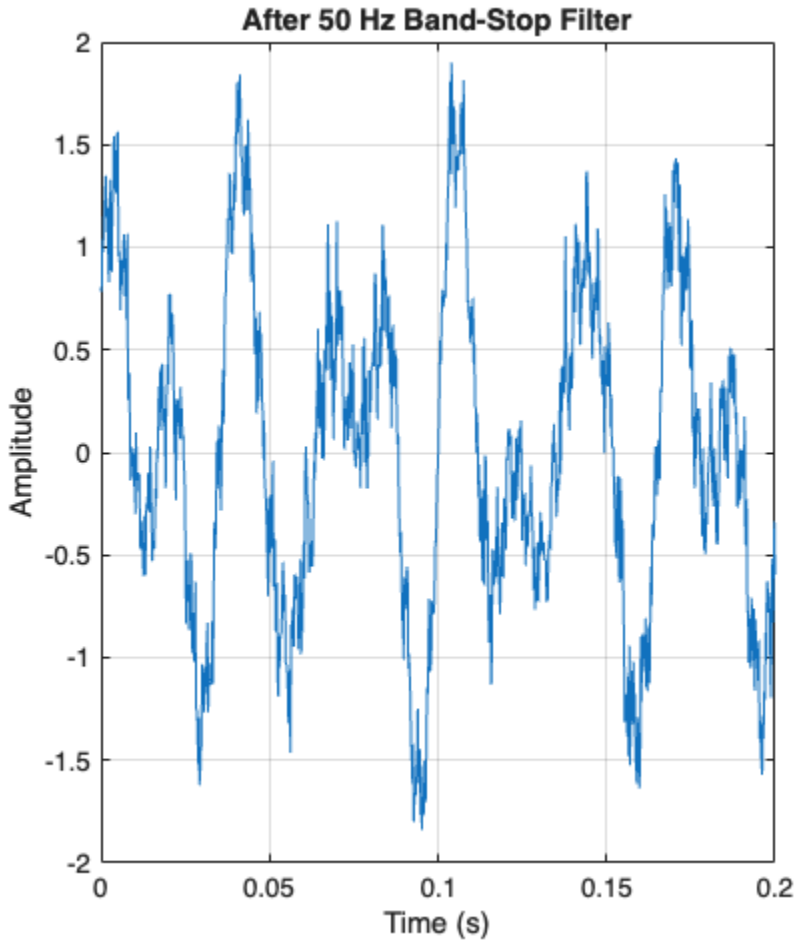
legend('Original Signal','Recieved Signal');

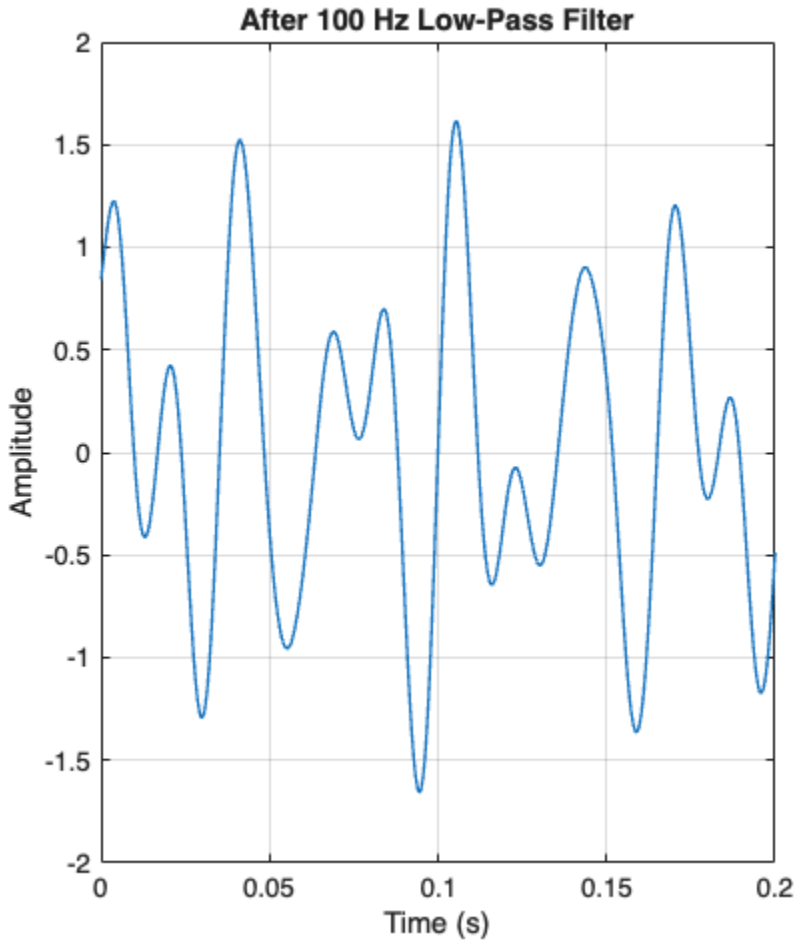
xlim([0,0.2]);
grid on;
hold off;


figure();
plot(t, notch_signal);
xlabel('Time (s)');
ylabel('Amplitude');
title('After 50 Hz Band-Stop Filter');
grid on;
xlim([0 0.2]);
```

```
figure();
plot(t, filtered_signal,'LineWidth', 1);
xlabel('Time (s)');
ylabel('Amplitude');
title('After 100 Hz Low-Pass Filter');
grid on;
xlim([0 0.2]);
```




After 50 Hz Band-Stop Filter
Amplitude
Time (s)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.05
0.1
0.15
0.2



SNR vs A for different H

sigma = 0.15

```
A_values = [0.5 0.75 1.0 1.25 1.5];
H_values = [0.2 0.3 0.4 0.5 0.6];
sigma_values = [0.05 0.10 0.15 0.20 0.25];
SNR_AH = zeros(length(A_values),length(H_values));

for i = 1:length(A_values)
    for j = 1:length(H_values)

        x_test = A_values(i)*sin(2*pi*f_rot*t) + ...
                 H_values(j)*sin(2*pi*2*f_rot*t);

        a_x = k*x_test;
        noise = 0.15*randn(size(t));

        SNR_AH(i,j) = ...
            10*log10(rms(a_x)^2/rms(noise)^2);
```

```
    end
end

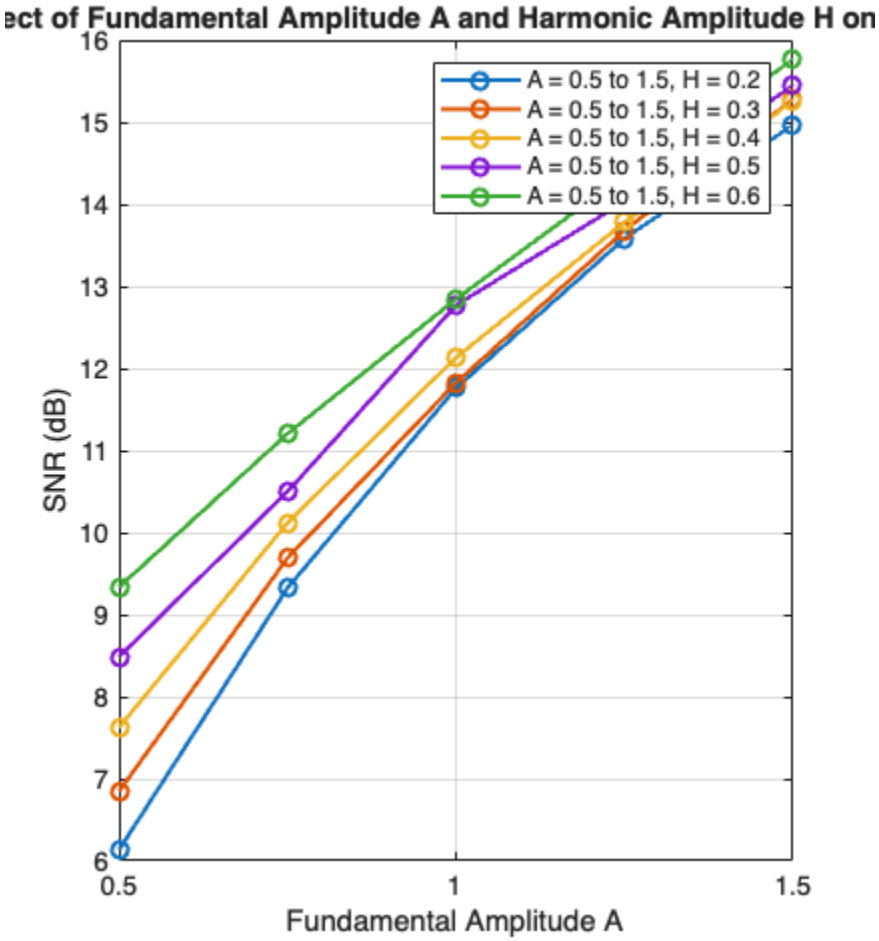
figure();

plot(A_values,SNR_AH(:,1),'o-','LineWidth',1.5);
hold on;
plot(A_values,SNR_AH(:,2),'o-','LineWidth',1.5);
plot(A_values,SNR_AH(:,3),'o-','LineWidth',1.5);
plot(A_values,SNR_AH(:,4),'o-','LineWidth',1.5);
plot(A_values,SNR_AH(:,5),'o-','LineWidth',1.5);

xlabel('Fundamental Amplitude A');
ylabel('SNR (dB)');
title('Effect of Fundamental Amplitude A and Harmonic Amplitude H on SNR');

legend('A = 0.5 to 1.5, H = 0.2', ...
       'A = 0.5 to 1.5, H = 0.3', ...
       'A = 0.5 to 1.5, H = 0.4', ...
       'A = 0.5 to 1.5, H = 0.5', ...
       'A = 0.5 to 1.5, H = 0.6');

grid on;
```



SNR vs H for different A

sigma = 0.15

```
SNR_AH = zeros(length(A_values),length(H_values));

for i = 1:length(A_values)
    for j = 1:length(H_values)

        x_test = A_values(i)*sin(2*pi*f_rot*t) + ...
            H_values(j)*sin(2*pi*2*f_rot*t);

        a_x = k*x_test;
        noise = 0.15*randn(size(t));

        SNR_AH(i,j) = ...
            10*log10(rms(a_x)^2/rms(noise)^2);

    end
end
```

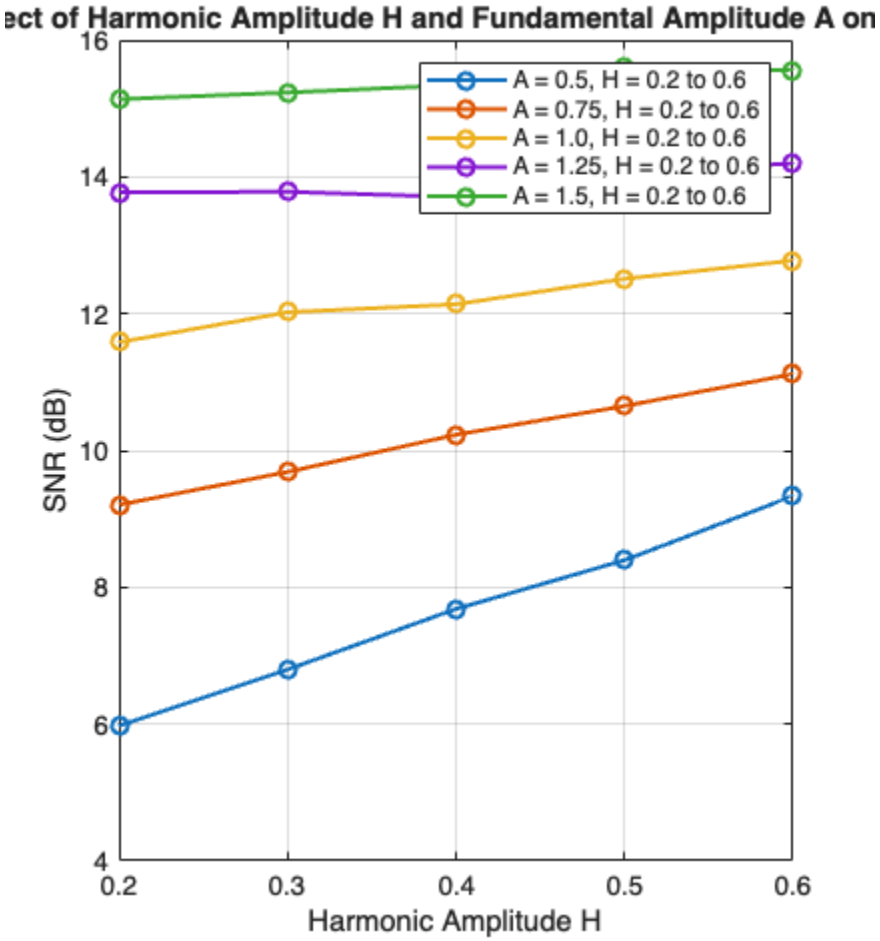
```
figure();

plot(H_values,SNR_AH(1,:),'o-','LineWidth',1.5);
hold on;
plot(H_values,SNR_AH(2,:),'o-','LineWidth',1.5);
plot(H_values,SNR_AH(3,:),'o-','LineWidth',1.5);
plot(H_values,SNR_AH(4,:),'o-','LineWidth',1.5);
plot(H_values,SNR_AH(5,:),'o-','LineWidth',1.5);

xlabel('Harmonic Amplitude H');
ylabel('SNR (dB)');
title('Effect of Harmonic Amplitude H and Fundamental Amplitude A on SNR');

legend('A = 0.5, H = 0.2 to 0.6', ...
       'A = 0.75, H = 0.2 to 0.6', ...
       'A = 1.0, H = 0.2 to 0.6', ...
       'A = 1.25, H = 0.2 to 0.6', ...
       'A = 1.5, H = 0.2 to 0.6');

grid on;
```



SNR vs sigma for different A

H = 0.4

```
SNR_Asigma = zeros(length(A_values),length(sigma_values));

for i = 1:length(A_values)
    for j = 1:length(sigma_values)

        x_test = A_values(i)*sin(2*pi*f_rot*t) + ...
                 0.4*sin(2*pi*2*f_rot*t);

        a_x = k*x_test;
        noise = sigma_values(j)*randn(size(t));

        SNR_Asigma(i,j) = ...
            10*log10(rms(a_x)^2/rms(noise)^2);

    end
end

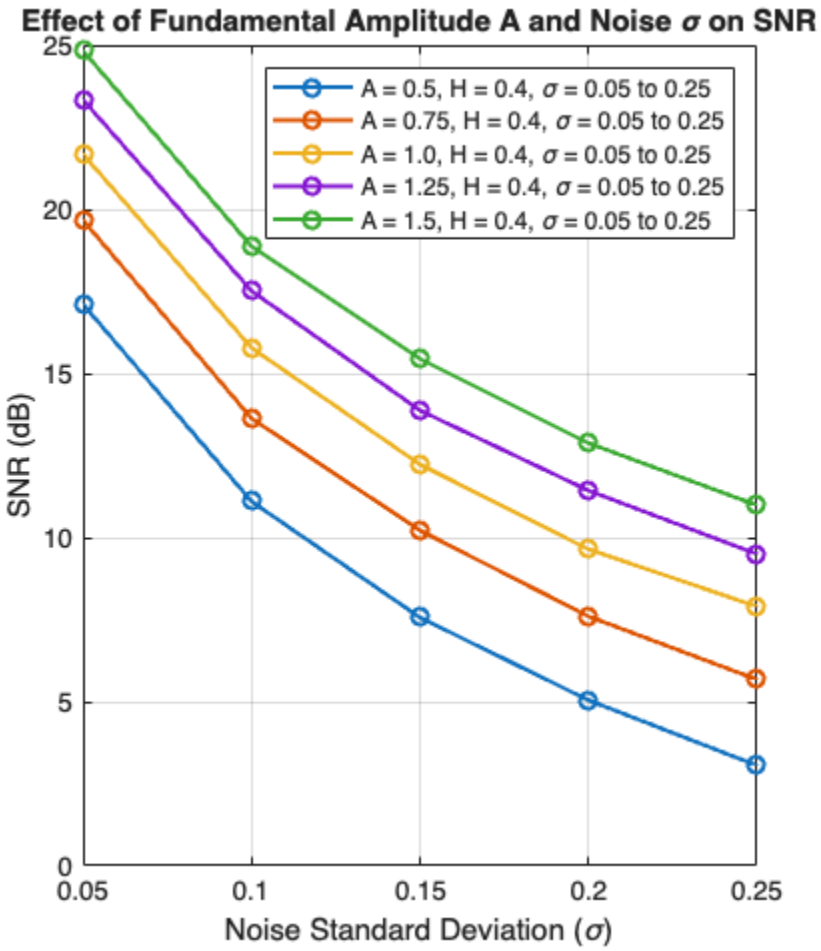
figure();

plot(sigma_values,SNR_Asigma(1,:),'o-','LineWidth',1.5);
hold on;
plot(sigma_values,SNR_Asigma(2,:),'o-','LineWidth',1.5);
plot(sigma_values,SNR_Asigma(3,:),'o-','LineWidth',1.5);
plot(sigma_values,SNR_Asigma(4,:),'o-','LineWidth',1.5);
plot(sigma_values,SNR_Asigma(5,:),'o-','LineWidth',1.5);

xlabel('Noise Standard Deviation (\sigma)');
ylabel('SNR (dB)');
title('Effect of Fundamental Amplitude A and Noise \sigma on SNR');

legend('A = 0.5, H = 0.4, \sigma = 0.05 to 0.25', ...
       'A = 0.75, H = 0.4, \sigma = 0.05 to 0.25', ...
       'A = 1.0, H = 0.4, \sigma = 0.05 to 0.25', ...
       'A = 1.25, H = 0.4, \sigma = 0.05 to 0.25', ...
       'A = 1.5, H = 0.4, \sigma = 0.05 to 0.25');

grid on;
```



Effect of Harmonic Amplitude H and Noise sigma on SNR

```
H_values = [0.2 0.3 0.4 0.5 0.6];
sigma_values = [0.05 0.10 0.15 0.20 0.25];

A_fixed = 1;
SNR_Hsigma = zeros(length(H_values),length(sigma_values));

for i = 1:length(H_values)

    for j = 1:length(sigma_values)

        % Project vibration signal
        x_test = A_fixed*sin(2*pi*f_rot*t) + ...
            H_values(i)*sin(2*pi*2*f_rot*t);

        % Attenuation
        a_x = k*x_test;
```

```
        % Random noise
        noise = sigma_values(j)*randn(size(t));

        % SNR
        SNR_Hsigma(i,j) = ...
            10*log10(rms(a_x)^2/rms(noise)^2);

    end

end
```

Plot - SNR vs sigma for different H

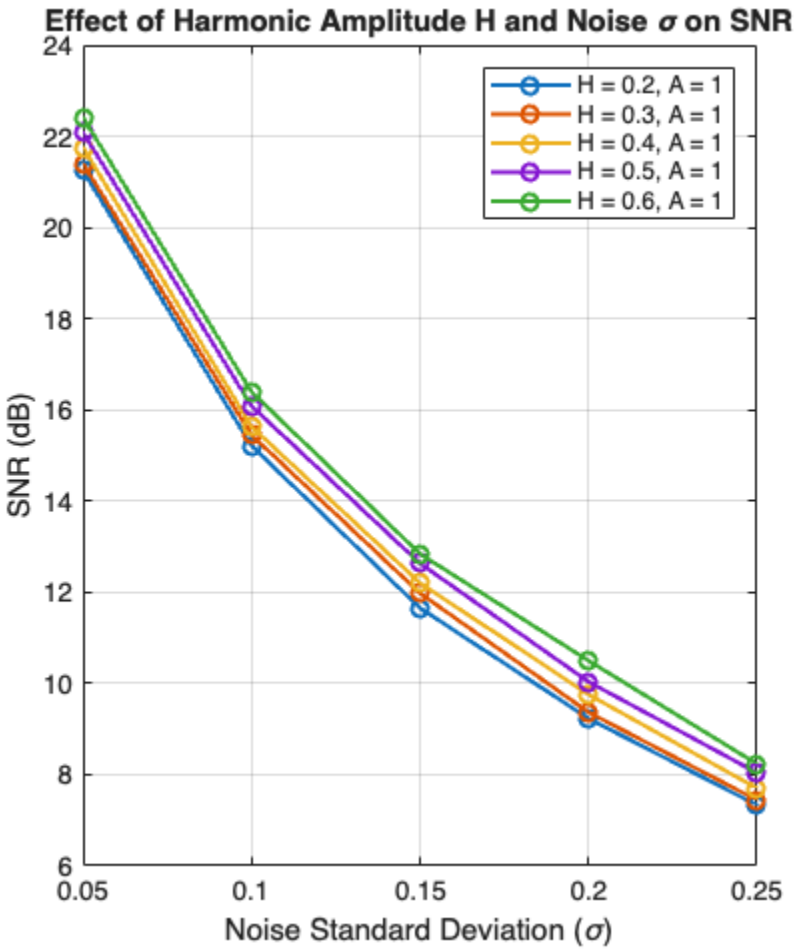
```
figure();

plot(sigma_values,SNR_Hsigma(1,:),'o-','LineWidth',1.5);
hold on;
plot(sigma_values,SNR_Hsigma(2,:),'o-','LineWidth',1.5);
plot(sigma_values,SNR_Hsigma(3,:),'o-','LineWidth',1.5);
plot(sigma_values,SNR_Hsigma(4,:),'o-','LineWidth',1.5);
plot(sigma_values,SNR_Hsigma(5,:),'o-','LineWidth',1.5);

xlabel('Noise Standard Deviation (\sigma)');
ylabel('SNR (dB)');
title('Effect of Harmonic Amplitude H and Noise \sigma on SNR');

legend('H = 0.2, A = 1', ...
    'H = 0.3, A = 1', ...
    'H = 0.4, A = 1', ...
    'H = 0.5, A = 1', ...
    'H = 0.6, A = 1');

grid on;
```



FFT for different motor speeds

```
RPM = [1500 1800 2100];

figure;
hold on;

for i = 1:length(RPM)

    f_rot = RPM(i)/60;

    x = A*sin(2*pi*f_rot*t) + ...
        H*sin(2*pi*2*f_rot*t);

    a_x = k*x;

    noise = A_electrical*sin(2*pi*f_electrical*t) + ...
        A_high*sin(2*pi*f_high*t) + ...
        sigma*randn(size(t));
```

```
    r_x = a_x + noise;

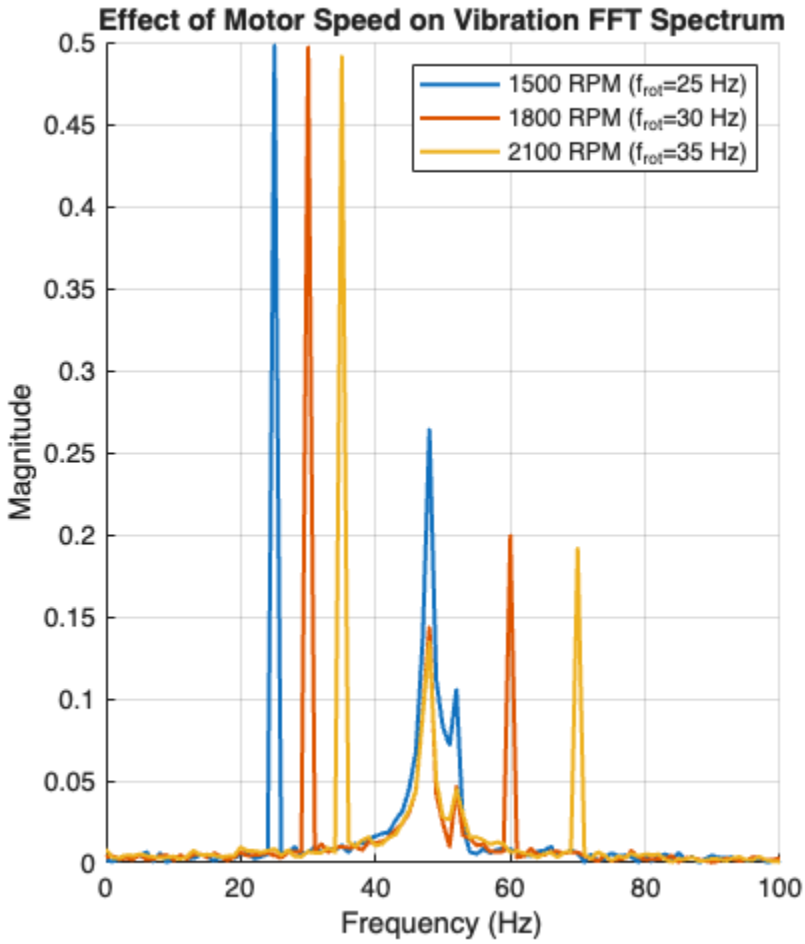
    notch_signal = filtfilt(bn,an,r_x);
    filtered_signal = filtfilt(bl,al,notch_signal);

    recovered_signal = filtered_signal/k;

    F = fft(recovered_signal);
    P = abs(F/L);

    plot(freq(1:floor(L/2)),P(1:floor(L/2)), ...
        'LineWidth',1.5, ...
        'DisplayName',sprintf('%d RPM (f_{rot}=%.0f Hz)',RPM(i),f_rot));
end

xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Effect of Motor Speed on Vibration FFT Spectrum');
legend('Location','northeast');
xlim([0 100]);
grid on;
hold off;
```



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